Engineering Applications of Artificial Intelligence 167 (2026) 113938

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Research paper

Bimodal temporal modeling reinforcement learning with safety mechanism for highway lane change in mixed traffic

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ARTICLE INFO

Keywords:
Highway lane change
Bimodal temporal modeling
Reinforcement learning
Safety mechanism
Mixed traffic

ABSTRACT

The differences in driving behavior between Connected and Autonomous Vehicles (CAVs) and Human-Driven Vehicles (HDVs) significantly impact traffic safety and efficiency in complex mixed environments. Ensuring that CAVs can safely and efficiently change lanes is a challenging issue. Previous research has focused on designing reward functions or enforcing basic constraints to ensure safety and efficiency. However, these methods often fail to model the interactions between vehicles in the observation space, limiting their ability to understand vehicle interactions and potential hazards. Additionally, biased sample selection can hinder understanding of temporal or historical trajectories, leading to instability in decision-making. To address these issues, we propose a bi-modal temporal modeling reinforcement learning algorithm combined with a lane-changing safety mechanism. This approach allows CAVs to model bi-modal information regarding observed states and vehicle relationships. The temporal module extracts representation from historical experiences while assessing collision risks during the lane-changing process, evaluating safety, and guiding the policy network to select optimal actions. We train and evaluate our approach using training rewards and average vehicle speed across three different traffic scenarios, testing the collision rate and average travel distance. The results demonstrate that our method significantly reduces collision rates and improves average vehicle speed in all three scenarios, proving its effectiveness.

1. Introduction

In recent decades, Connected and Autonomous Vehicles (CAVs) have emerged as a crucial technology in the transportation sector, with their automation and perception capabilities promising improvements in traffic flow efficiency, congestion reduction, and road safety (Cao et al., 2022). However, in scenarios where CAVs coexist with Human-Driven Vehicles (HDVs), the subjective decision-making of human drivers often prioritizes minimizing travel costs over overall efficiency. Differences in driving behavior, dynamic traffic changes, and other factors can increase the complexity and uncertainty of traffic. Lane-changing behavior, in particular, plays a significant role in highway safety and traffic flow. While many previous methods have relied on fixed rules or assumptions, such approaches fail to adapt to the dynamic and complex nature of mixed traffic environments. Lane-changing behavior, in particular, plays a significant role in highway safety and traffic flow, as a large proportion of highway collisions and traffic disruptions are directly related to unsafe or poorly coordinated lane-changing maneuvers. However, in mixed traffic environments where CAVs coexist with Human-Driven Vehicles (HDVs), the inherent uncertainty and heterogeneity of human driving behaviors significantly increase the complexity of lane-changing decision-making. Many existing approaches rely on fixed rules or simplified assumptions, which limits their adaptability to such dynamic and interactive traffic conditions. Lane changing is a critical aspect of vehicle behavior, as many highway accidents and congestion issues arise from unsafe or incorrect lane-changing operations (Ali et al., 2020). Consequently, there is growing attention on the safety and efficiency of CAVs during lane changes. To ensure that CAVs can change lanes safely and efficiently, it is essential for them not only to make decisions regarding objects on the road but also to consider the reactions of other vehicles in the target lane and potential interactions. Unlike traditional rule-based methods, which struggle with adaptability, our proposed approach integrates both temporal modeling and vehicle relationship dynamics, allowing

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https://doi.org/10.1016/j.engappai.2026.113938
Received 14 October 2024; Received in revised form 22 December 2025; Accepted 20 January 2026
Available online 27 January 2026
0952-1976/© 2026 Published by Elsevier Ltd.

CAVs to better respond to changing traffic conditions. Proper lane-changing behavior is crucial for ensuring smooth traffic flow, reducing collision rates, and minimizing travel time. Therefore, modeling the behavior and impact of CAVs in mixed traffic flow, as well as exploring travel mechanisms and control methods, is vital.

Previous CAV lane-changing methods, including rule-based models (Cheng et al., 2023b), optimization-based models (Liu et al., 2023), and game-theoretic models (Cheng et al., 2023a), often lack adaptability and learning capabilities in complex, dynamic traffic scenarios. These methods typically rely on predefined rules or assumptions, failing to capture the subtle interactions and uncertainties present in real-world driving environments. As a result, they struggle to improve performance and adapt to constantly changing traffic conditions. To overcome these limitations, deep reinforcement learning (DRL) offers lane-changing models, such as deep Q-network (Wang et al., 2023), advantage actor-critic (Moghaddasi et al., 2024), and proximal policy optimization (Huang and Qu, 2023), which excel in learning efficiency, sample efficiency, and stability. These models learn optimal lane-changing strategies through interactions with their environment, making them suitable for a variety of applications and environments. Despite carefully designed reward functions and behavioral constraints, state information is crucial for determining an agent's decisions and rewards. Current methods rely solely on observed state information, lacking the ability to model vehicle relationships. This limits their effectiveness in capturing dynamic changes and potential dangers within complex traffic dynamics. Furthermore, while experience replay can enhance sample utilization and leverage historical data, it suffers from sample selection bias and often overlooks temporal dependencies, leading to unstable model decisions. Additionally, constraint-based methods during lane changing typically depend on static rules and policies, neglecting the reactions and potential interactions of other vehicles in the target lane, which undermines safety during lane-changing operations.

To address the limitations mentioned above, we propose a novel bi-modal temporal modeling reinforcement learning (RL) algorithm, enhanced with a lane-changing safety mechanism. Unlike traditional rule-based or optimization-based approaches, which rely on static assumptions and struggle in dynamic environments, our method explicitly integrates temporal dependencies and inter-vehicle relationship dynamics. We employ a temporal module to account for time dependencies and construct corresponding policy and critic networks that output discrete policy probabilities and value functions. Compared with constraint-based safety rules, our game-theoretic safety mechanism dynamically adapts to surrounding vehicles' strategies, allowing the model to anticipate potential conflicts and ensure collision-free maneuvers. This integrated design reduces collision risks, improves decision stability, and enhances overall traffic safety.

To bridge the gap between simulation-based evaluation and real-world applicability, the adopted traffic environment is designed to approximate realistic highway mixed traffic conditions. Specifically, the Intelligent Driver Model (IDM) and LC2013 lane-changing model have been extensively validated in prior studies for reproducing human driving behaviors under varying traffic densities and driving styles. By incorporating heterogeneous vehicle compositions, multiple traffic flow levels, and diverse driving aggressiveness, the simulation environment serves as a controlled and partially representative proxy for real-world highway mixed traffic scenarios, capturing key interaction patterns while abstracting away sensing uncertainty and infrastructure-level constraints, enabling systematic and controlled evaluation of lane-changing safety and efficiency.

Compared to previous methods, our proposed bi-modal temporal modeling reinforcement learning approach with an integrated safety mechanism offers several key advantages. By modeling both the environmental states and vehicle interactions, the agent can better detect potential hazards and adapt to complex traffic dynamics. The temporal module captures time dependencies and historical traffic trajectories, enabling the agent to develop more robust and accurate lane-changing strategies. Furthermore, the integrated safety mechanism prioritizes safe lane changes based on real-time traffic conditions and vehicle dynamics, reducing collision risks and enhancing overall traffic safety.

The aim of this paper is to develop a lane-changing decision model that promotes safer and more efficient lane-changing behaviors for CAVs in mixed traffic environments. We utilize the Simulation of Urban Mobility (SUMO) (Lopez et al., 2018) to simulate the lane-changing benefits and evolution processes of both HDVs and CAVs in mixed traffic. We analyze the impact of different traffic patterns based on training rewards and average vehicle speeds. Through experimental evaluations, we assess the collision rate and average travel distance of our approach, comparing it with related research to demonstrate its effectiveness. The main contributions of this paper can be summarized as follows:

(1) We have modeled the observation state of CAVs and inter-vehicle relationships in a bi-modal manner, encoding different modal data and extracting features to combine their representation into a unified feature vector for decision prediction.
(2) To enhance the understanding of temporal and historical trajectory information, we have modeled and stored feature vectors with temporal context, using historical experience to guide the policy network in selecting optimal actions.
(3) To account for vehicle behavior and interactions during lane changes, we have incorporated a safety mechanism that prioritizes safe operations based on surrounding traffic conditions and vehicle dynamics, thereby enhancing overall safety.

Next, in Section 2, we will introduce related work, identify research gaps, and discuss the main contributions of this study. In Section 3, we will describe the bi-modal temporal modeling reinforcement learning approach and the associated safety mechanism. Section 4 will evaluate the proposed model on the SUMO-based platform. In Section 5, we will discuss the significance of this study and assess the strengths and limitations of our results. Finally, we will provide the conclusion and prospects for future work in Section 6.

2. Literature review

The lane-changing behavior of vehicles is crucial for driving, significantly impacting traffic flow and road safety. Many traffic accidents and congestion on highways result from improper lane-changing operations, leading to increased attention on the issue of safe lane changing (Monteiro and Ioannou, 2023). Various simulation methods have been developed for highway lane-changing behavior, focusing primarily on two problems: lane-changing decision-making (Dai et al., 2023) and capturing the dynamic process of lane changing while quantifying its impact on traffic flow (Hamedi et al., 2023). These strategies can be broadly categorized into rule-based, optimization-based, game-based, and learning-based models.

Rule-based models define lane changes based on specific criteria, such as maintaining a safe distance and considering turn signals, providing straightforward actions (Aksjonov and Kyrki, 2023; Chala and Kóczy, 2024). For example, el abidine Kherroubi et al. (2021) proposes various rules and uses a gap acceptance algorithm to determine vehicle behavior before merging into the target lane. However, rule-based methods lack adaptability to unexpected situations due to their oversimplified representation of real-world scenarios, and they fail to capture the subtle decision-making processes of drivers. In contrast, optimization-based methods optimize lane change decisions using mathematical frameworks such as dynamic programming or evolutionary algorithms, incorporating vehicle interactions such as acceleration and steering angles (Zhou et al., 2023). Research has focused on cooperative merging algorithms for automated vehicles and HDVs, employing Model Predictive Control (MPC) to generate smooth trajectories for triplets of vehicles with varying compositions (Karimi et al.,

2020). Particularly in lane-changing scenarios, MPC-based methods have demonstrated good performance; however, they heavily rely on accurate dynamic merging models (Aramrattana et al., 2020). Safety has typically been addressed through rule-based or constraint-based methods, ensuring collision avoidance but often lacking adaptability (Ye et al., 2020; Ji and Levinson, 2020a). Recent developments in robust and uncertainty-aware safe RL (He et al., 2024; Lu et al., 2025) have sought to provide formal guarantees or adaptive safety mechanisms under uncertainty. Similarly, new game-theoretic models for mandatory and merging lane changes (Cheng et al., 2023a; Qu et al., 2025) have extended the classical frameworks with empirical validation in mixed traffic. These advances demonstrate that integrating adaptability with theoretical grounding is a promising direction, which aligns with the game-theoretic safety controller proposed in this study. On the other hand, game-theoretic models conceptualize lane changing as a strategic interaction between drivers, aiming to optimize individual objectives while balancing cooperation and competition (Von Neumann and Morgenstern, 2007; Zhou et al., 2023). While these methods extend traditional theory to address incomplete information and dynamic scenarios, they may encounter challenges in coordinating individual and collective interests, leading to suboptimal outcomes or instability.

Finally, learning-based models leverage the representational power of neural networks to capture complex patterns and correlations in driving behavior from data. In the context of autonomous vehicle control, these models can be divided into Supervised Learning and Reinforcement Learning (Pavel et al., 2022). Supervised Learning trains models on labeled real-world data to replicate expert driving performance, but it heavily depends on the quality and coverage of the training data (Budach et al., 2022). In contrast, Reinforcement Learning, particularly Deep Reinforcement Learning (DRL), has achieved significant success in recent years (Dang et al., 2023). By combining reinforcement learning with deep neural networks, these models enable autonomous vehicles to perform lane change operations in multi-lane highway scenarios with surrounding traffic (Ye et al., 2020; Hu et al., 2020). For instance, policies can be trained using Proximal Policy Optimization (PPO) to allow vehicles to execute lane changes on crowded highways (Ye et al., 2020). Another example involves an agent using the Deep Q-Network (DQN) algorithm to perform single-lane lane changes on highways (Hu et al., 2020). Additionally, the Double Deep Q-Network (DDQN) algorithm, combined with a human lane change decision model as a safety supervisor, trains the agent to wait for a suitable gap before changing lanes and to maintain its lane afterward (Chen et al., 2020). In Wang et al. (2019), the Deep Deterministic Policy Gradient (DDPG) is employed to train an agent for lane changes on a three-lane highway, simulating entering and exiting vehicles using the Intelligent Driver Model (IDM) (Bai et al., 2024). More recently, transformer-based reinforcement learning has been proposed to better capture long-range temporal dependencies and multi-agent interactions (Parisotto et al., 2020; Chen et al., 2021). Building on these pioneering studies, recent surveys and extensions (Li et al., 2023; Zhuang et al., 2024) have systematically reviewed transformer architectures in RL and demonstrated improvements in robustness and offline training efficiency through variants such as Robust Decision Transformer and Reinformer. These works highlight the increasing importance of sequence modeling approaches in both online and offline RL, providing a stronger foundation for our bimodal temporal modeling framework. These models demonstrate superior capability in capturing long-range temporal dependencies compared to traditional RNN-based approaches. In addition, multi-agent reinforcement learning (MARL) has been increasingly explored in traffic domains, enabling cooperative and competitive interactions among vehicles (Chu et al., 2019; Hua et al., 2025). More recent surveys (Zhang et al., 2024; Yadav et al., 2023) have provided comprehensive reviews of MARL in autonomous driving, summarizing progress on scalability, communication mechanisms, and real-world applicability. These works emphasize the potential of MARL for complex urban and highway scenarios, while also highlighting challenges such as training stability and policy transfer, which motivate our choice to focus first on a single-agent PPO framework with an external safety mechanism. Such methods show promise for scaling autonomous driving systems to dense traffic environments. Our work complements these advances by integrating game-theoretic safety constraints into the RL framework, highlighting the importance of balancing learning-based policies with explicit safety guarantees.

When applying reinforcement learning to CAVs, a common goal is to design reward functions that balance safety and efficiency, aiming to avoid collisions, minimize traffic disruption, and achieve smoother acceleration (Hu et al., 2019). Careful selection of action spaces is also crucial, encompassing continuous acceleration control (Li et al., 2021), discrete maneuver selection (Quang Tran and Bae, 2020), and high-level decision-making (Triest et al., 2020). Given the partial observability of the environment, RL agents often rely on local observations from onboard sensors. To address this, researchers have proposed recursive strategies (Bouton et al., 2018) and attention mechanisms (Chen et al., 2019) to infer unobserved information, such as driver intent, from observed history. Additionally, multi-agent RL can facilitate collaborative perception among networked vehicles (Kaushik et al., 2019). These methods emphasize integrating safety and efficiency through carefully designed reward functions. Studies like Ren et al. (2020) and Bouton et al. (2020) address safety by enforcing basic constraints during training or using separate modules to identify and mitigate unsafe actions (Li et al., 2021). Furthermore, Ye et al. (2020) and Hwang et al. (2022) evaluate candidate actions and substitute unsafe actions with safer alternatives. Alternatively, Chen et al. (2023) proposes a priority-based safety monitor to enhance CAV safety during highway merging, while Hu et al. (2020) embeds safety directly into the policy network through masking or priority-based coordination.

In this study, we utilized the microscopic traffic simulation software (SUMO) to control HDVs within the simulated environment. The longitudinal motion of HDVs was modeled using the Intelligent Driver Model (IDM), while lateral motion was controlled based on the LC2013 lane change model. The datasets includes a wide range of traffic conditions, such as varying vehicle densities (1920 pcu/h, 2400 pcu/h, and 3000 pcu/h) and different driving styles (aggressive, normal, cautious), ensuring sufficient diversity for robust RL model training. It is important to note that our RL agents were not provided with any prior information about vehicle interactions or the underlying system logic.

Additionally, we introduced a novel lane-change safety mechanism to assess the safety of autonomous vehicle lane-changing behavior. We validated our approach across various traffic scenarios of differing complexity and demonstrated its effectiveness through collision rate and average travel distance evaluations. Our results were also compared with related work to highlight the improvements in safety and performance.

3. Methodology

3.1. Problem formulation

When vehicles need to avoid collisions or overtake, they typically perform lane-changing maneuvers. Our objective is to enable CAVs to gradually execute safe and efficient lane changes within dynamic traffic environments through self-learning. In addition to maximizing the direct rewards from lane-changing maneuvers, the goal is to account for vehicle interactions and assess the safety of lane-changing in terms of collision risk. We consider a simulated four-lane highway scenario where a CAV needs to perform a lane change, as illustrated in Fig. 1. We used Proximal Policy Optimization (PPO) (Huang and Qu, 2023) to train the CAV to execute safe and efficient lane-changing behaviors. In the simulated environment, we observed the CAV's behavior and optimized its lane-changing decisions through reinforcement learning. The experimental results demonstrate the effectiveness of PPO in improving both the safety and efficiency of CAV lane-changing, offering an effective solution for CAV lane-change behavior.

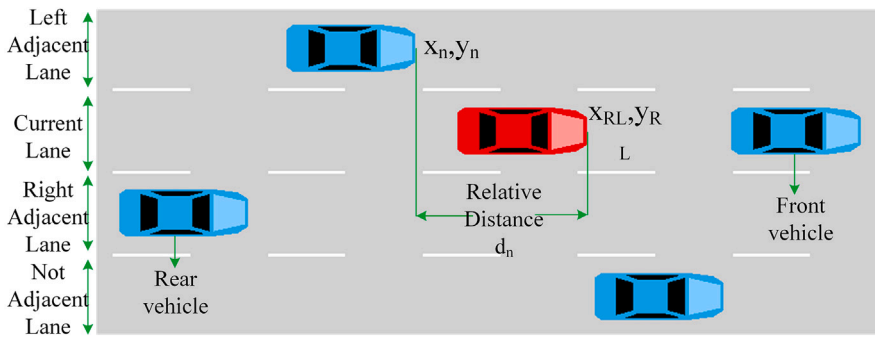

Fig. 1. Traffic environment of the RL agent consists of four lanes and multiple other front and rear vehicles. CAVs(red) and HDVs(blue) are randomly generated in the lanes.

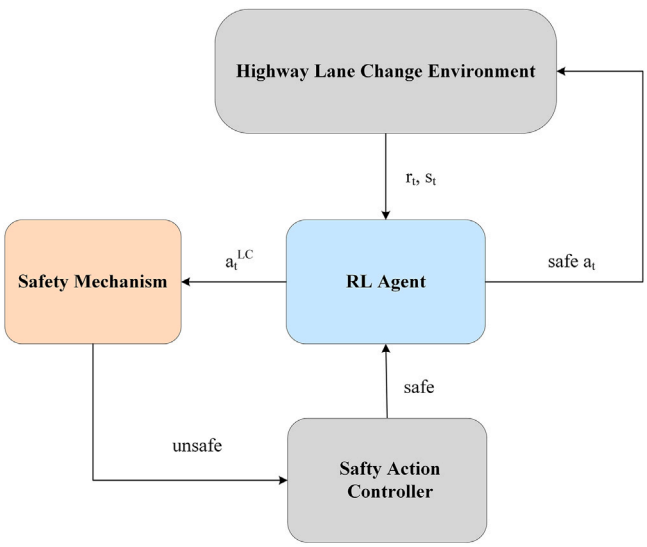

Fig. 2. Based on the current time step t and state s_t, the RL agent execute the action a_t in the environment and receives a reward r_t and a new state s_{t+1}.

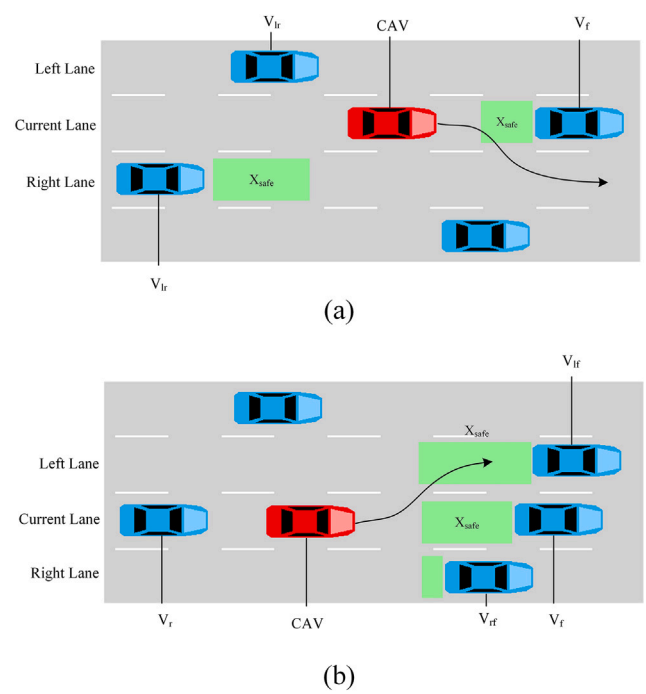

Fig. 3. (a) Visualization of a lane change maneuver to the right. (b) Visualization of a lane change maneuver to the right. Safety distances are illustrated by green areas.

In this study, we utilized the microscopic traffic simulation software (SUMO) to control HDVs within the simulated environment. The longitudinal motion of HDVs was modeled using the Intelligent Driver Model (IDM) (Bai et al., 2024), while lateral motion was controlled based on the LC2013 lane change model. It is important to note that our RL agents were not provided with any prior information about vehicle interactions or the underlying system logic.

3.1.1. General problem definition

We model the highway lane-changing scenario as a Markov Decision Process (MDP) (Howard, 1960). An MDP is defined by the tuple (S, A, P, R, ρ_0, γ, T), where S is the set of states, A is the set of actions, $P(s'|s,a)$ is the transition function that describes the system's behavior, and $R(s,a,s')$ is the immediate reward function, defined as $r : S \times A \rightarrow R(s,a,s')$. ρ_0 defines the initial state distribution, γ is the discount factor (ranging from 0 to 1), and T represents the time horizon. In reinforcement learning, the objective is to find an optimal policy $\pi^* : S \rightarrow A$ that maximizes the expected cumulative return for each state.

As shown in Fig. 2, the RL agent and the environment interact in a sequence of discrete time steps $t = 0, 1, 2, \ldots$. At each time step t, the RL agent observes the current state $s_t \in S$ of the environment and selects an action $a_t \in A$. After executing action a_t, the agent receives a reward $r_t \in R$ and transitions to a new state s_{t+1}. The policy π is a mapping from S to A, assigning an action to each state. The objective of the RL agent, from a long-term perspective, is to find a policy that maximizes the cumulative return. The general form of the loss function for updating the RL policy (Ye et al., 2020) is:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\right] \tag{1}$$

Here, ε is a hyperparameter, and $r_t(\theta)$ represents the probability ratio, defined as $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{old}}(a_t|s_t)$. In this method, the probability ratio $r_t(\theta)$ is clipped within the range $(1-\varepsilon, 1+\varepsilon)$ and multiplied by the advantage estimate $\hat{A}_t$. The clipping helps stabilize updates by ensuring that large policy changes are avoided, depending on whether the advantage is positive or negative. This results in the final value L^{CLIP} of the clipped objective function. The complete RL algorithm utilizes fixed-length episodes of length T in each iteration, where the policy is run for T time steps. The agent gathers data at each time step, using generalized advantage estimation to improve learning efficiency.

$$\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \cdots + \cdots + (\gamma\lambda)^{T-t+1}\delta_{T-1} \tag{2}$$

where γ is the discount factor. Then the RL algorithm optimizes the data of these time steps with small batches for K epochs.

As shown in Fig. 3, when the RL agent prepares to perform a lane-changing maneuver, ensuring the safety of the lane change is

critical. To achieve this, we implement a safety mechanism that assesses potential dangers before proceeding. Unlike previous studies that relied solely on position-based classifications, we incorporate varying safety distances, X_{safe} based on the speed and acceleration of the surrounding vehicles. Specifically, we calculate the minimum longitudinal safety distance X_{safe} in the forward direction by summing the vehicle's current speed v_t multiplied by the driver's minimum reaction time, plus half of the product of the vehicle's maximum deceleration and the square of the reaction time. The effective safety margin decreases as the maneuver risk increases within this range.

When the distance to the target vehicle is sufficiently large, the lane change does not require game-theoretic involvement. As illustrated in Fig. 3, when the vehicle is within the safe distance threshold, the lane change also does not need to engage in a game. Therefore, we define the lane-changing game entry mechanism based on the safety distance threshold (Ji and Levinson, 2020a) as follows:

$$\log_{\text{dis}} = \begin{cases} 1, & 0 < x_{\text{RL}} - x_{\text{target}} - l < X_{\text{safe}}, \\ 0, & \text{Otherwise.} \end{cases} \tag{3}$$

where t is the time when the position of the CAV, x_{RL} and the target vehicle x_{target} are observed. l represents the length of the CAV, and X_{safe} denotes the minimum safe distance required for the CAV. If the safety distance threshold is satisfied, then $\log_{\text{dis}} = 1$ indicating that there is a risk for lane changing. In this case, the CAV must engage in a game with the target vehicle to evaluate the safety of the maneuver.

Additionally, driving style refers to characteristic behavioral patterns of drivers, which significantly influence vehicle motion in highway environments, which determine the vehicle's motion characteristics. Different driving styles manifest distinct driving features that influence the dynamic performance of vehicles. In highway environments, driving styles are generally categorized into conservative, normal, and aggressive models, each significantly impacting lane-changing decisions. Specifically, conservative drivers typically maintain lower speeds and reduce lane-changing frequency, prioritizing safety. Normal drivers seek a balance between safety and efficiency, exhibiting moderate speeds and lane-changing rates. In contrast, aggressive drivers tend to drive at higher speeds and change lanes frequently to maximize travel efficiency.

To simulate the impact of different driving styles on autonomous vehicles, we assume that the longitudinal control of Human-Driven Vehicles (HDVs) follows the Intelligent Driver Model (IDM) (Bai et al., 2024), which incorporates desired speed, inter-vehicle distance, and acceleration/deceleration behaviors influenced by varying driving habits. Additionally, the Minimize Overall Braking Induced by Lane Changes (MOBIL) (Khelfa and Tordeux, 2021) model is employed for lateral control. This model uses vehicle acceleration as an input variable and can be integrated with most car-following models. The expression for determining the longitudinal acceleration of the kth HDV in the IDM is as follows:

$$a_k = a_{\max}\left[1 - \left(\frac{v_k}{v_0}\right)^{\delta} - \left(\frac{s^*(v_k, \Delta v_k)}{s_k}\right)^2\right] \tag{4}$$

where v_k, s_k, δ, and v_0 denote the speed, actual gap, acceleration exponent, approach rate, and desired speed of the kth HDV, respectively. The desired minimum gap for the kth HDV is given by:

$$s^*(v_k, \Delta v_k) = s_0 + v_k T + \frac{v_k \Delta v_k}{2\sqrt{a_{\max} b}} \tag{5}$$

where T, s_0, $a_{\max}$ and b represent the safe time headway, minimum distance, maximum comfortable acceleration, and deceleration, respectively.

3.1.2. System architecture

The overall system architecture is illustrated in Fig. 4. The system comprises three main components: the simulation environment, the RL Agent, and the safety mechanism. The simulation environment includes road networks, traffic conditions, and various challenging scenarios, which are generated using SUMO to facilitate the training of the RL Agent through interaction. The RL Agent is trained using the Proximal Policy Optimization (PPO) algorithm, enabling the CAV to learn high-level decision-making strategies while interacting with the surrounding traffic environment. Additionally, the safety mechanism evaluates the safety of lane-changing actions taken by the RL Agent. In cases where a lane-changing action is deemed dangerous, a game-theoretic algorithm is utilized to model the lane-changing interaction as a non-cooperative game between two entities, determining a safe course of action.

3.1.3. State description

In the scenario illustrated in Fig. 1, the intelligent agent has limited visibility, focusing on its own lane and the preceding and following vehicles in the adjacent two lanes. The observed state s is composed of two key components. The first component captures the agent's own information: lateral position x_{RL}, longitudinal position y_{RL}, velocity v_{RL}, acceleration a_{RL}, the number of surrounding vehicles n_{RL}, vehicle density d_{RL}, and the average speed v_{L} of the road.

The second component includes relationship information regarding surrounding vehicles, defined as those within a longitudinal distance of 100 m in both the current and adjacent lanes. These are the nearest vehicles to the ego vehicle and can range from only leading and following vehicles to a maximum of eight observed vehicles. As shown in Fig. 1, the state of each surrounding vehicle is represented by its lateral position x_n, longitudinal position y_n, velocity v_n, acceleration a_n, and a binary indicator F_n that signals whether the vehicle is within the observable vicinity of the ego vehicle. Additionally, the relative distances d between the intelligent agent and the surrounding vehicles are captured, integrating the information from both components into the overall state representation.

3.1.4. Agent action

An action represents the response of an agent to its environment. In this context, we define a discrete action space, denoted as $a_t \in 0, 1, 2, 3, 4$, which consists of five possible actions: idle, turning left, turning right, accelerating, and decelerating.

3.1.5. Reward function and payoff function

This article aims to guide the intelligent agent in learning an optimal strategy that balances safety, efficiency, and comfort in driving. To achieve this, we employ a multi-objective reward function, which is defined as a linear weighted function:

$$r = w_c r_c + w_s r_s + w_l r_l \tag{6}$$

In our framework, the multi-objective reward function is designed to balance safety, stability, and comfort during lane-changing maneuvers for Connected and Automated Vehicles (CAVs). The relative importance of each factor is captured through the weight parameters w_c, w_s, and w_l, which correspond to collision assessment (r_c), stability assessment (r_s), and lane change cost assessment (r_l), respectively. The weights w_c, w_s, and w_l were tuned through a grid search over candidate values in the range [0,1], evaluated on simulation episodes covering diverse traffic densities and lane-changing scenarios. The tuning criterion was the maximization of the cumulative episode reward while ensuring no significant increase in collision rate. The final configuration was set to $w_c = 0.7$, $w_s = 0.2$, and $w_l = 0.1$. Sensitivity analysis confirmed the robustness of our method within ±10% variations in these weights, demonstrating that the model's performance is stable across a reasonable range of parameter values.

Given that safety is the primary concern in vehicular decision-making, a higher weight w_c is assigned to the collision assessment. This ensures that the agent prioritizes avoiding collisions above all else,

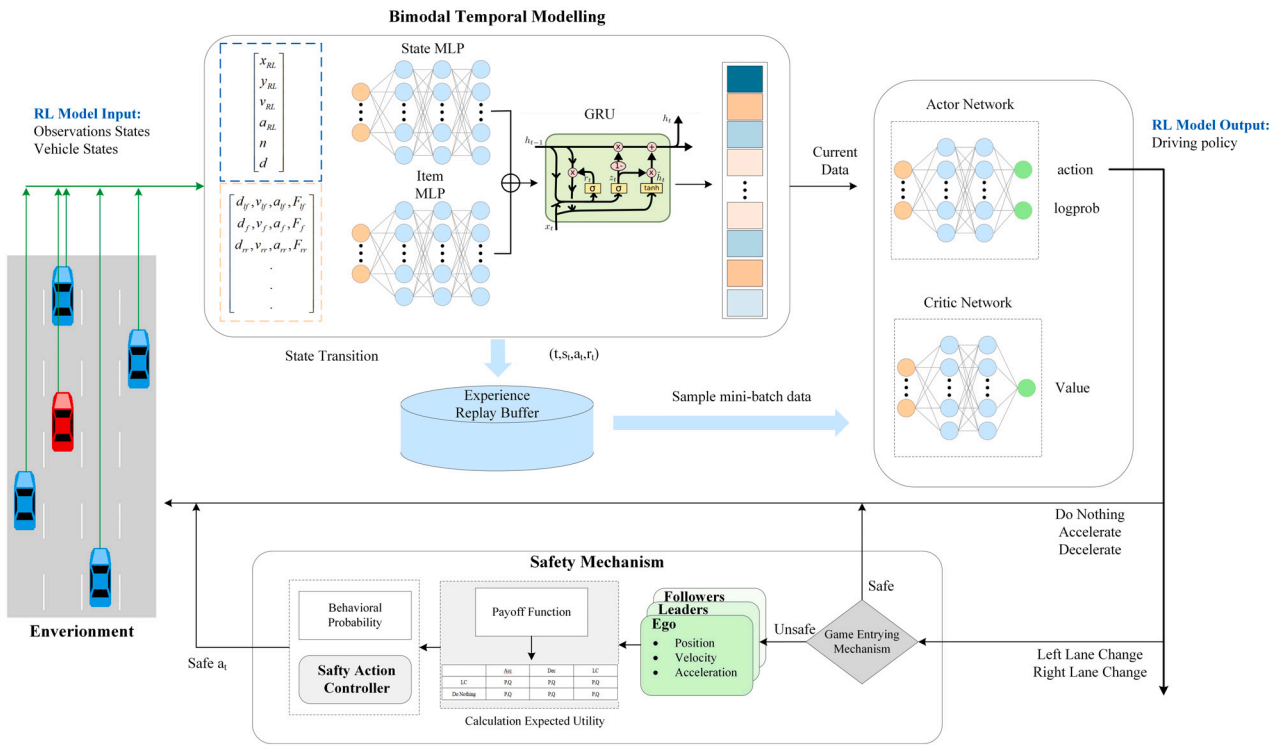

Fig. 4. The proposed system architecture for bimodal temporal modeling reinforcement learning with safety mechanism.

reflecting real-world imperatives where the consequences of accidents can be severe.

$$r_c = \begin{cases} -10, & \text{If collision happened,} \\ 0, & \text{Otherwise.} \end{cases} \tag{7}$$

r_c is defined with strict penalties, assigning a score of −10 for any collision event and 0 otherwise. This stark contrast emphasizes that safety breaches are heavily discouraged in the learning process, guiding the agent toward cautious behavior.

The stability assessment focuses on maintaining optimal speeds, which is critical for ensuring smooth traffic flow and reducing the risk of abrupt maneuvers that could lead to accidents. This weight is set lower than w_c, reflecting the importance of safety over speed but still ensuring that the agent values efficiency in its operation. Therefore, stability assessment r_s is defined as:

$$r_s = \begin{cases} \frac{v_t - v_{\min}}{v_{\max} - v_{\min}}, & \text{If } v_t < v_{\min}, \\ \frac{v_{\max} - v_t}{v_{\max} - v_{\min}}, & \text{If } v_t > v_{\max}, \\ 0, & \text{Otherwise .} \end{cases} \tag{8}$$

where v_t is the current speed of the vehicle, and $v_{\min}$ and $v_{\max}$ are the minimum and maximum speeds of the vehicle, respectively. The minimum speed $v_{\min}$ and maximum speed $v_{\max}$ represent the safe operational speed limits for the vehicle. $v_{\min}$ can be set based on traffic regulations or safety standards, often aligned with the lowest permissible speed on a given roadway, typically around 30 km/h or local speed limits. $v_{\max}$ should reflect the speed limit of the road, which could be obtained from traffic sign data or predefined settings in the simulation environment. These values can also be adjusted based on factors such as road conditions and vehicle capabilities. The stability reward is calculated based on how closely the current speed v_t aligns with predefined minimum $v_{\min}$ and maximum speeds $v_{\max}$. This function provides a continuous incentive for the agent to operate within safe speed limits, promoting both safety and efficiency.

In order to maintain driving comfort, the lane change cost is penalized to discourage unnecessary lane changes that could disrupt traffic flow and increase the likelihood of accidents. This weight is lower than both w_c and w_s, reflecting the notion that while comfort and smoothness are important, they should not come at the expense of safety. Therefore, the cost evaluation of lane changes is defined as:

$$r_l = \begin{cases} -1, & \text{if a lane change occurs,} \\ 0, & \text{Otherwise.} \end{cases} \tag{9}$$

The lane change cost incurs a penalty of −1 for each lane change, promoting a more stable and less erratic driving style. By doing so, the model encourages the agent to evaluate the necessity of each maneuver carefully.

The weighted reward function integrates these factors into a cohesive decision-making framework, guiding the intelligent agent toward optimal behavior in complex traffic scenarios. The emphasis on safety through the high weight on collision assessment reflects real-world priorities, where the prevention of accidents is paramount. The higher weight for w_c over w_s indicates a deliberate design choice that places safety first. This prioritization ensures that even in scenarios where speed might be sacrificed, the agent remains focused on avoiding collisions. However, the significant weight on w_s also illustrates the need for maintaining reasonable speed, which is critical for overall traffic efficiency. While lane changes are essential for maneuvering through traffic, the lower weight on lane change cost assessment acknowledges that frequent lane changes can disrupt flow and lead to safety issues. By applying a mild penalty, the model promotes smoother driving patterns without entirely eliminating the agent's ability to change lanes when necessary.

To improve both safety and efficiency in the current reward function, the focus should be on better balancing the conflicting objectives of avoiding collisions while maintaining optimal speeds and minimizing unnecessary lane changes. This would help maintain consistency between the high-level objectives and the detailed decision-making process. At the same time, when implementing the lane change safety mechanism, it is necessary to focus on the safety payoff function of vehicles to ensure lane change safety. The decision-making effectiveness of a CAV's lane change strategy—i.e., that of ego vehicles—depends

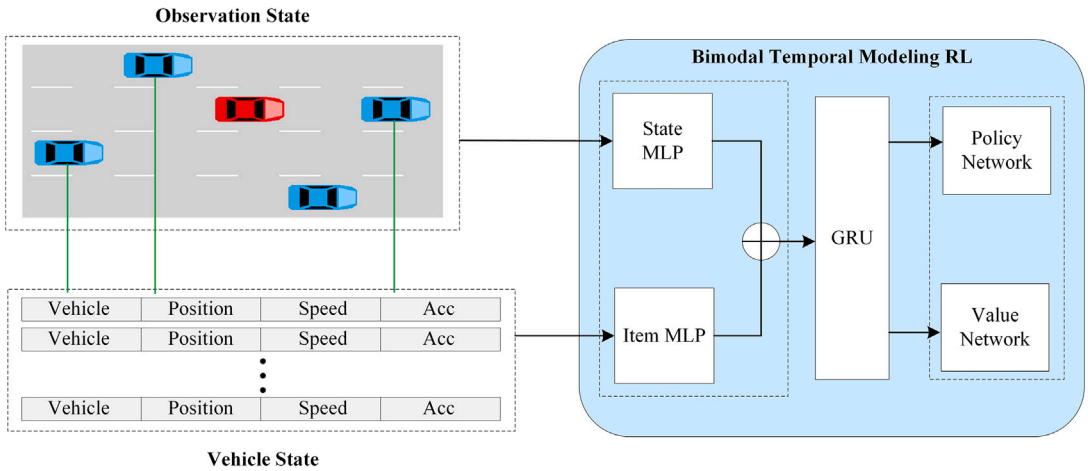

Fig. 5. Framework of bimodal temporal modeling, with two modals state on the left and state encoding as well as representational modeling on the right.

on the expected actions of conflicting vehicles. The payoff function adopted in this paper is as follows:

$$U_n = c_s(f_{f,s}(a,v) + f_{r,s}(a,v)) + c_e f_e(v) \tag{10}$$

where c_s and v_e are the weight coefficients of the safety and efficiency payoff functions. $f_{f,s}(a,v)$ and $f_{r,s}(a,v)$ are the safety payoff functions of the vehicle in front and behind, respectively, and $f_e(v)$ is the efficiency payoff function of the vehicle n. We predict the time to collision (TTC) based on instantaneous velocities and accelerations of interacting vehicles. Specifically, the safety payoff function is defined as:

$$f_s(a,v,t) = \frac{v_r - v_f}{a_f - a_r} - \frac{\left((v_f - v_r)^2 - 2(a_f - a_r)(x_f - x_r - l_f)\right)^{\frac{1}{2}}}{a_f - a_r} \tag{11}$$

where v_f and v_r are the velocities of the front and rear vehicles at time t, a_f and a_r are their accelerations at time t, x_f and x_r are their longitudinal positions, and l_r is the length of the rear vehicle. This formulation estimates TTC by solving the kinematic equations of motion under constant accelerations, thereby quantifying collision risk in real time. It has been adapted from standard collision-time formulations (Ji and Levinson, 2020b). While paying attention to driving safety, drivers also need to consider driving efficiency. The efficiency payoff function is defined as:

$$f_e(v,t) = \frac{v_n}{v_{\max}} + 1 - e^{-\frac{1}{v_n - v_{\text{best}}}} \tag{12}$$

where, v_n, $v_{\max}$ and v_{best} are respectively the speed of the vehicle at the moment t, the maximum speed of the vehicle and the optimal speed expected of the vehicle.

3.2. Bimodal temporal modeling RL

We apply the PPO method (Ye et al., 2020), which uses a clipping-based agent loss function, and combine the policy agent with the value function error term. However, we provide prior knowledge to obtain richer representation information through bi-modal temporal modeling and integrate the lane change safety mechanism to output more secure discrete strategies and value functions. Algorithms 1 and 2 describe our bi-modal temporal modeling reinforcement learning approach and lane change safety mechanism.

As shown in Fig. 5, the process of reinforcement learning in bi-modal temporal modeling can be divided into three stages:

(1) Collect the bi-modal information of the agent regarding the observed state and the vehicle state, as outlined in line 14 of Algorithm 1.

(2) State coding and representation modeling of bi-modal information are performed.

We use two independent multilayer perceptrons (MLPs) to encode the ego state and the inter-vehicle relationship state, respectively. Each MLP consists of two hidden layers with 64 and 64 neurons, each followed by ReLU activation. The output of the ego-state encoder is a 64-dimensional feature vector, and the output of the relationship-state encoder is also a 64-dimensional feature vector. These two vectors are concatenated into a 128-dimensional joint representation, which is then passed through a fully connected projection layer to obtain a 64-dimensional fused vector. This fused feature vector is subsequently fed into the temporal module and policy/value networks. Such a design ensures that the ego-state and relational information are first processed independently, preserving their unique properties, and then combined to provide a richer representation for decision prediction.

Algorithm 1 Bimodal Temporal Modeling RL

1: Initialize replay buffers RB
2: Initialize hyperparameters, hidden state vector h and gate vectors r, z
3: **for** e = 1 to N_{episode} **do**
4: Initialize environment and get initial state s, vehicle state vs
5: **for** t = 1 to N_{maxstep} **do**
6: Sample action a_t by policy net
7: **if** a_t is unsafe by Eq. (3) **then**
8: Store unsafe experience (t, s_t, vs_t, a_t, r_t) in RB
9: a_t = Compute a safe action by Safety Mechanism(**Algorithm 2**)
10: **end if**
11: Execute safe action a_t, observe r_t, s_{t+1} and vs_{t+1} according to Eq. (6)
12: Set $s_t = s_{t+1}$ and $vs_t = vs_{t+1}$
13: Store experience (t, s_t, vs_t, a_t, r_t) in RB
14: **end for**
15: **for** k = 1 to N_{epochs} **do**
16: Reset the hidden state h
17: Sample mini-batches of experiences from RB
18: Update h_{t-1} to h_t
19: Update policy net and value net
20: **end for**
21: **end for**

(3) Capture time dependencies using time-series modules.

We introduce a gated recurrent unit (GRU) to capture temporal dependencies. The GRU is chosen for its balance of expressive power and computational efficiency compared to other recurrent architectures such as LSTM. It incorporates a memory unit that enables the agent to

effectively store and retrieve historical information. It incorporates a memory unit that enables the agent to effectively store and retrieve historical information.

In lane-changing scenarios, vehicles must consider not only the current environment but also recent traffic conditions and prior decisions. By maintaining a hidden state, the GRU helps the vehicle anticipate potential conflicts earlier and improves decision stability in rapidly changing traffic. Formally, as shown in line 19 of Algorithm 1, the hidden state $\tilde{h}t$ of the current time step is updated as $ht = (1-z_t)\odot h_{t-1} + z_t \odot \tilde{h}_t$, where z_t is the update gate and $\tilde{h}_t$ is the candidate hidden state. This formulation allows the GRU to capture long-term dependencies while mitigating gradient vanishing or explosion during training. While introducing the GRU does incur additional computational cost (about 6–8 ms per decision step in our setup), it significantly improves safety and robustness. To mitigate runtime concerns, lightweight optimization techniques such as model pruning and parallelization can be applied in future deployments.

(4) Construct the corresponding policy and critic networks to apply the learned policy to a specific problem.

During the training process of RL agents, each step needs to be recorded. Unlike other algorithms that collect a set of experience transfer samples (s_t, a_t, r_t, s_{t+1}), we collect the experience (t, s_t, vs_t, a_t, r_t) corresponding to each time step t, as shown in line 14 of Algorithm 1. Since we are considering time series, each time step is treated as sequence data rather than just a single state–action pair. These sequences include state sequences, action sequences, and reward sequences, meaning that the entire sequence needs to be stored. After obtaining the sequence data, we improve the policy by optimizing the objective function using gradient ascent. In each training iteration, we extract mini-batches from the replay buffer and replay the entire sequence data rather than individual experiences, as shown in line 18 of Algorithm 1. The advantage estimate is calculated to update the policy gradient and improve the strategy. This process is repeated until a predefined number of training sessions is completed.

3.3. Safety mechanism

In order to account for the behavior and interaction between vehicles during CAV lane changes, we incorporate a safety mechanism based on game theory. This mechanism evaluates the safe lane-change operation according to surrounding traffic conditions and vehicle dynamics, thus improving safety. The proposed game-theory-based safety mechanism is outlined in Algorithm 2.

As shown in Fig. 6, the game-based safety mechanism consists of three stages:

(1) The first stage determines whether it is necessary to engage in the game by calculating the minimum safe distance from surrounding vehicles. Based on this, it assesses whether a game should be played with other vehicles using Eq. (3), as shown in Algorithm 2 (Line 4). If the game is not required, the action is directly returned for the agent to perform (Lines 5–7).

(2) In the second stage, the payoff for the vehicles involved in the game is calculated based on both safety and efficiency, through the agent's perception of the surrounding environment. If there is a safety risk in the lane-changing maneuver, the payoff function for the vehicles participating in the game is calculated using Eq. (10) (Line 8) to determine the safety actions.

(3) In the third stage, the Nash equilibrium of the game problem is solved to calculate the optimal strategy and the probability of strategy selection. The game-theoretic Nash equilibrium is computed based on the positions and velocities of the vehicles involved. Simultaneously, the safety action controller outputs actions deemed safe by the safety mechanism, while restricting unsafe actions. The proposed safety mechanism belongs to a finite strategy game (Fudenberg and Tirole, 1991), which guarantees at least one pure strategy or mixed strategy Nash equilibrium, as shown in Algorithm 2 (Line 9).

Algorithm 2 Game-based Safety Mechanism

1: **Input:** Current lane change action a_t, Current vehicle position, speed and acceleration x_i, v_i, a_i of all vehicles
2: **Output:** Safe lane change action a_{safe}
3: Initialize payoff functions $U_i(t)$ of vehicles
4: Determine the game-vehicle and non-game-vehicle by Eq. (3)
5: **if** non-game-vehicle **then**
6: Execute action a_t
7: **end if**
8: Calculate expected payoff matrix A, B of two vehicles by Eq. (10)
9: Find Nash equilibrium strategies $u_i(s_1, s_2, ..., s_n)$ by Eq. (13) and Eq. (14)
10: **if** PureStrategiesExist **then**
11: Execute pure strategy
12: **else**
13: Calculate max mixed strategy probabilities (p^*, q_1^*, q_2^*) by Eq. (17)
14: Execute action based on mixed strategy probabilities
15: **end if**

In our implementation, the computation of the two-vehicle game reaches equilibrium within a few milliseconds per decision step, which is acceptable for real-time deployment. As the number of interacting vehicles increases, the size of the payoff matrices grows and the computational complexity may rise exponentially. To address this, approximate Nash solvers or hierarchical decomposition strategies can be used to maintain tractability in multi-vehicle interactions. Therefore, while the current implementation is efficient for highway lane-change scenarios with limited participants, future extensions will incorporate scalable algorithms to ensure feasibility in dense urban traffic.

If a Nash equilibrium is found (Line 9), the best action is returned (Line 11). During the game, each participant has different strategy sets, such as $S_n = s_1, s_2, \dots, s_n$ for the ego vehicle and $T_n = t_1, t_2, \dots, t_n$ for the target vehicle. If there exists a combination of actions that prevents any participant from unilaterally changing their control actions to obtain a higher payoff U_i, satisfying the following formula, this combination is defined as a Nash equilibrium:

$$u_i(s_1, s_2, \dots, s_n) > u_i(s_1, s_2, \dots, s_n), \quad \forall s_n \in S_n \tag{13}$$

$$u_i(t_1, t_2, \dots, t_n) > u_i(t_1, t_2, \dots, t_n), \quad \forall t_n \in T_n \tag{14}$$

When there is no pure strategy Nash equilibrium, the payoff expectation of mixed strategies can be calculated based on the different probabilities of the strategies. In this study, for example, Algorithm 2 (Lines 13–14) calculates the mixed strategy using the formula and returns the action with the highest probability (Lines 13–14). In our system, the game-theoretic safety controller is invoked only when the headway-based safety check signals a potential conflict. The resulting game is a fixed 2-player normal-form game: the ego CAV chooses between lane-change, no-change while the most critical HDV in the target lane chooses among accelerate, decelerate, keep, i.e., a 2×3 payoff matrix computed from the safety/efficiency terms Eq. (10). We first test for a pure-strategy Nash equilibrium; if none exists, we compute the mixed equilibrium in closed form for the 2×3 case. This keeps the per-step complexity O(1) with a small constant overhead. For scenes with more vehicles, we maintain a 2-player abstraction by selecting the single most critical opponent (minimum safety margin). A top-K extension is straightforward — solving up to K independent 2×3 games — leading to linear O(K) scaling with negligible additional latency per game. The expected profit coefficient of the mixed strategy reflects the expectation and belief about other drivers' decisions. Here, p_{lc} and $p_{donotlc}$ represent the probabilities that a vehicle will change lanes or not change lanes, while q_{acc}, q_{dec}, and q_{lc} represent the probabilities that the target vehicle will accelerate, decelerate, or change lanes, respectively.

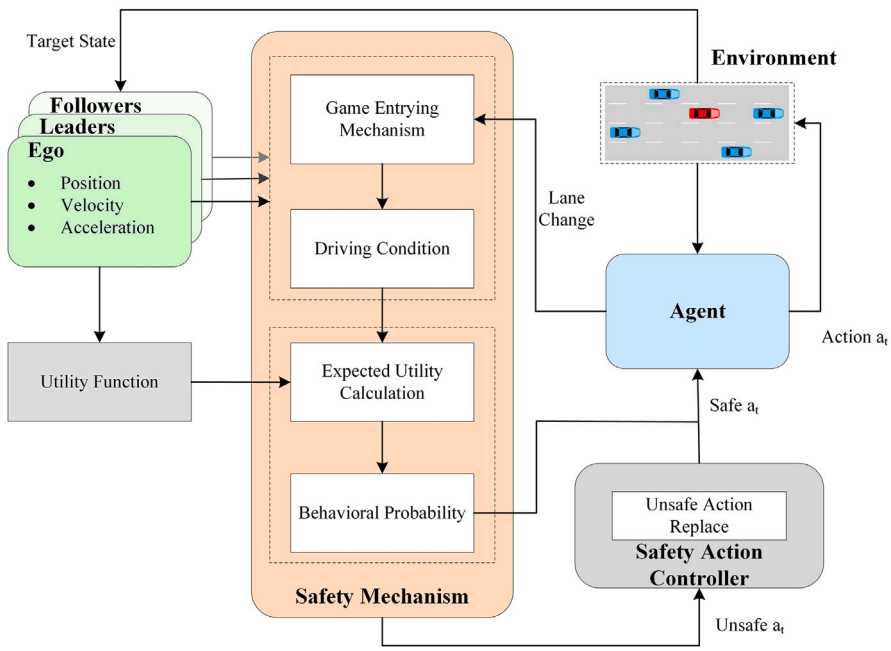

Fig. 6. Framework of game-based safety mechanisms.

These expectations are probabilities between 0 and 1. Specifically, the expected returns, E_{RL} and E_{target}, of the mixed strategies for the ego vehicle and the target vehicle are calculated as the sum of the products of the corresponding probabilities and the returns of each mixed strategy:

$$E_{\mathrm{RL}}(p, q_1, q_2) = \sum_{i=1,2}^{n} p_i\Big(\sum_{j=1,2,3}^{m} q_j U_{ij}\Big) \tag{15}$$

$$E_{\mathrm{target}}(p, q_1, q_2) = \sum_{i=1,2}^{n} p_i\Big(\sum_{j=1,2,3}^{m} q_j U_{ij}\Big) \tag{16}$$

where $p_1 = p_{lc}$, $p_2 = p_{donotlc}$, $q_1 = q_{acc}$, $q_2 = q_{dec}$, and $q_3 = q_{lc}$, In this set of equations, the optimal solution can be obtained by maximizing the expected returns of the main vehicle and the target vehicle, Specifically, to find the strategy with the maximum return, the formula $E_{\mathrm{RL}}(p^*, q_1^*, q_2^*) \geq E_{\mathrm{RL}}(p^*, q_1^*, q_2^*)$ satisfying the maximum return strategy of the main vehicle and the formula $E_{\mathrm{target}}(p^*, q_1^*, q_2^*) \geq E_{\mathrm{target}}(p^*, q_1^*, q_2^*)$ satisfying the maximum return strategy of the target vehicle. To further find the maximum strategy combination (p^*, q_1^*, q_2^*), solve for (p^*, q_1^*, q_2^*) under the maximum return according to the following formula:

$$\begin{cases} p_1 = \frac{\partial E(p,q_1,q_2)}{\partial q_1}, \\ p_2 = \frac{\partial E(p,q_1,q_2)}{\partial q_2}, \\ q = \frac{\partial E(p,q_1,q_2)}{\partial p}. \end{cases} \tag{17}$$

4. Experiments and performance evaluations

4.1. Simulation setup

In this part, we introduce the simulation environment and the design of the highway lane-change scenario. We then compare the proposed approach to several state-of-the-art baseline methods under different traffic patterns.

In this paper, we adopted the open-source micro-traffic simulation software Simulation of Urban MObility (SUMO) (Lopez et al., 2018) to conduct simulation research on the proposed bi-modal temporal modeling enhancement method and game-based safety mechanism. To increase the complexity of the experimental environment, we defined three different traffic modes: easy, medium, and difficult. These modes are characterized by different numbers of initial HDVs on the lane. Additionally, the driving style of HDVs is a factor that cannot be ignored. CAVs need to adjust the lane-change benefits according to the driving style of HDVs. For example, as shown in Eq. (10), normal drivers tend to weigh the benefits of both indicators, while aggressive drivers prioritize driving efficiency and may ignore safety costs to some extent. In contrast, careful drivers are more concerned about driving safety. Our simulation parameters for modeling conservative, ordinary, and aggressive longitudinal driving behavior are provided in Table 1. The settings for the simulation scenario and training parameters are shown in Table 2.

Table 1
Simulated parameter estimates for modeling aggressive, ordinary and cautious behavior in our scenarios.

	Parameter	Aggressive	Normal	Cautious
IDM	d_0	2.0 m	2.5 m	3.0 m
	Acc_{max}	5.0 m/s^2	4.0 m/s^2	3.0 m/s^2
	Acc_{min}	7.5 m/s^2	6.0 m/s^2	5.0 m/s^2
MOBIL	p	0.0	0.3	1.0
	a_{th}	0.0 m/s^2	0.1 m/s^2	0.4 m/s^2
	b_{safe}	7.5 m/s^2	6.0 m/s^2	5.0 m/s^2

Table 2
Simulation and training hyper-parameters.

Parameter	Value
Total lane length	7000 m
$N_{episode}$	5000
Batch size	16
Learning rate	0.0005
Simulation time step	0.6 s
Easy traffic mode	1920 pcu/h
Medium traffic mode	2400 pcu/h
Hard traffic mode	3000 pcu/h

In this experiment, we randomly select different departure positions, initial speeds, and departure lanes in the highway simulation environment. We input vehicle flow rates of 1920, 2400, and 3000 vehicles per hour according to the easy, medium, and difficult traffic modes,

respectively, to ensure that there are enough vehicles entering and passing through the highway.

The performance of our bi-modal temporal modeling reinforcement learning algorithm heavily relies on the careful selection and tuning of parameters and hyperparameters. We employed a systematic approach to optimize these elements, ensuring robustness and efficiency in lane-changing behavior. Initial values for parameters, such as learning rates and discount factors, were chosen based on established benchmarks in the literature, typically ranging from 0.0001 to 0.01 for learning rates. The discount factor was set to 0.99 to balance immediate and future rewards. To explore a predefined range of hyperparameters, we tested common values, including batch size, the number of hidden layers, and the number of neurons per layer. Batch sizes ranged from 32 to 256, and hidden layer configurations spanned from one to three layers with neuron counts between 64 and 512. Both the actor network and the critic network of the proposed model used a neural network structure that included a fully connected layer, a GRU layer, and another fully connected layer. The dimensions of the input matrix are set to [3, 500, 29], where 3 represents the historical points, 500 is the length of an episode, and 29 indicates the traffic state. To optimize the learning process, we employed the Adam optimizer, known for its adaptive learning rate capabilities, allowing for faster convergence compared to traditional stochastic gradient descent methods. Additionally, we adjusted the learning rate dynamically using a learning rate scheduler, reducing the rate after a plateau in performance was observed during training. Metrics such as average reward, collision rates, and average travel distance were used to evaluate the model's effectiveness during the training and validation phases. Throughout the training process, we conducted iterative evaluations, adjusting hyperparameters based on observed performance improvements. The model underwent 5000 iterative simulation rounds, with a maximum of 500 simulation steps in each round. When the number of training samples stored by the model exceeded 16, 16 data samples were collected. The strategy network and value network were iterated 15 times, with 3 samples sent for training each time. This adaptive approach facilitated a deeper understanding of how each parameter influenced the model's behavior. These metrics guided our tuning process, helping to refine hyperparameter settings for optimal performance. The other comparison algorithms involved in this chapter remained consistent, except for differences in neural network structure, training parameter settings, simulation environment settings, etc., which ensures the reliability of the experimental comparison results.

4.2. Ramp merging scenario

To provide an initial indication of the transferability of the proposed framework to a mandatory merging scenario, an additional highway on-ramp merging scenario was constructed in the SUMO simulation environment. As shown in Table 3, the scenario consists of a three-lane highway mainline and a single-lane on-ramp that merges into the rightmost lane of the mainline through a priority-based junction. This configuration emulates a typical real-world highway on-ramp, where merging vehicles are required to yield to mainline traffic, thereby introducing mandatory merging behavior and intensified vehicle interactions. The mainline is divided into three consecutive segments with lengths of 400 m, resulting in a total length of approximately 1.2 km. The on-ramp has a length of approximately 150 m, which is sufficient to induce realistic merging pressure while avoiding excessive acceleration space.

The merging junction adopts a priority control rule, where mainline vehicles have higher priority and ramp vehicles must yield. This setting is consistent with common highway traffic regulations and creates challenging interaction conditions for autonomous lane-changing decisions. traffic flow is considered in this scenario, including background human-driven vehicles (HDVs) and controlled autonomous vehicles (CAVs). HDVs are modeled using the Intelligent Driver Model (IDM) for longitudinal motion, while their lateral behavior follows the same lane-changing model as in the mainline experiments. The CAV employs the proposed decision-making framework for lateral maneuvers, while longitudinal control is handled by the same car-following model to ensure consistency across scenarios. The traffic demand is set to a moderate level to ensure frequent interactions without causing severe congestion. The mainline flow rate is set to 2400 vehicles per hour, while the ramp flow rate is set to 600 vehicles per hour. The speed limits are 33.33 m/s (120 km/h) for the mainline and 22.22 m/s (80 km/h) for the on-ramp. All other simulation parameters remain consistent with those described in Section 4.1 to ensure a fair evaluation.

Table 3
Key parameters of the ramp merging scenario.

Parameter	Value
Mainline lanes	3
On-ramp lanes	1
Mainline speed limit	33.33 m/s
On-ramp speed limit	22.22 m/s
Mainline flow rate	2400 pcu/h
Ramp flow rate	600 pcu/h
Simulation duration	3600 s

Fig. 7. Correlation between traffic density and driving style effects on average road vehicle speeds.

It should be noted that this ramp merging scenario is introduced as a preliminary generalization test rather than a dedicated optimization of ramp merging strategies. The policy trained in the mainline lane-changing scenario is directly applied to the ramp merging scenario without additional fine-tuning, allowing for an objective assessment of the robustness and transferability of the proposed framework.

4.3. Experimental results

In this part, we evaluate the effectiveness of the proposed bi-modal temporal modeling reinforcement method and the safety mechanism. Our comprehensive experiment consists of two main parts. In the first part, we assess the effectiveness of the proposed bi-modal temporal modeling reinforcement method and the game-based safety mechanism by analyzing training rewards and average speeds. In the second part, we compare the performance of our approach to three mainstream baseline methods across three different highway scenarios.

In addition to the mainline lane-changing experiments, a supplementary ramp merging experiment is conducted to assess the generalization capability of the proposed framework under a mandatory merging scenario.

We studied the performance of the proposed methods in terms of safety and efficiency under different traffic densities and driving styles. To measure safety, we calculated the percentage of collisions that occurred and examined the effects of traffic density and driving style

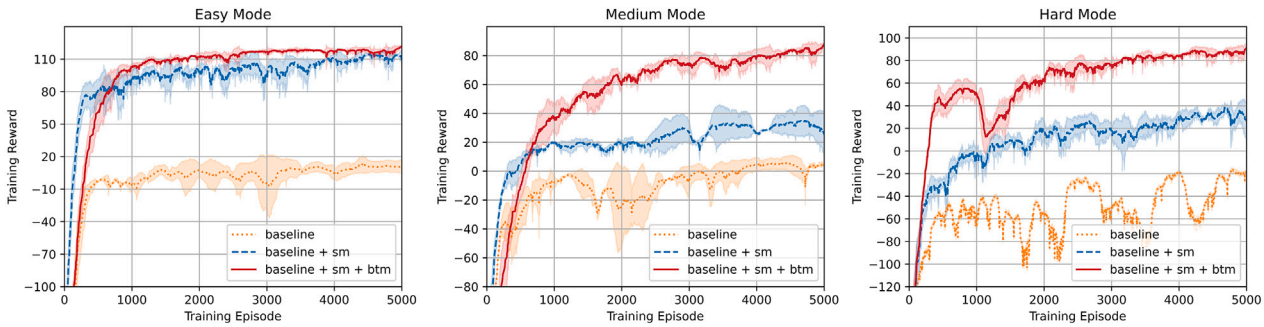

Fig. 8. Training reward comparisons in three traffic modes.

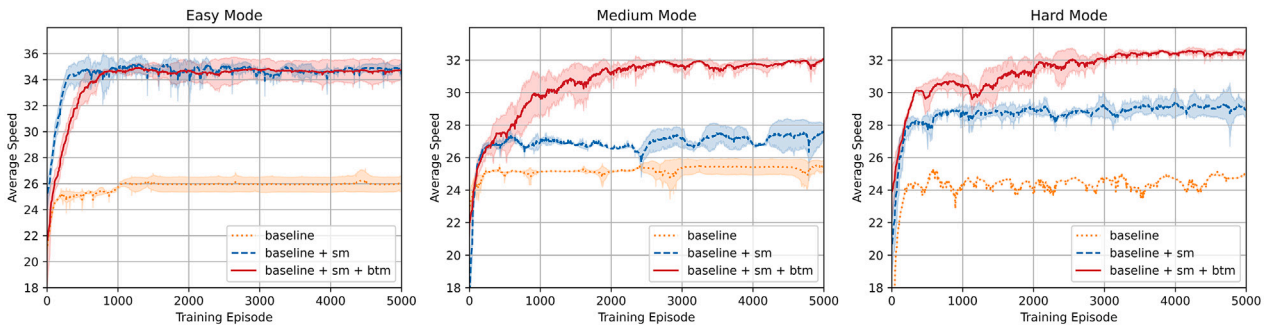

Fig. 9. Average speed comparisons in three traffic modes.

on CAVs. To measure efficiency, we used the average distance traveled by vehicles and the average driving speed. Focusing on scenarios with HDVs, we considered three driving style scenarios for HDVs, as well as three hybrid driving style scenarios. Fig. 7 illustrates the influence of traffic difficulty and driving style on the average speed of vehicles, demonstrating that both factors affect driving efficiency. As traffic difficulty increases and the aggression of HDVs intensifies, the average speed of road vehicles decreases.

As shown in Figs. 8 and 9, we trained the proposed bi-modal temporal modeling method and the game-based safety mechanism under different traffic modes to obtain the training reward and average driving speed for each turn of the CAV. It is evident from Fig. 7 that the driving performance of the CAV has improved. The safety mechanism has enhanced both the training reward and average speed, significantly outperforming the baseline model (i.e., PPO without the safety mechanism). Furthermore, the bi-modal temporal modeling method has led to additional improvements in training reward and average speed, demonstrating the validity of both the bi-modal temporal modeling reinforcement method and the game-based safety mechanism.

Specifically, in Fig. 8, the approach with the safety mechanism achieved a higher training reward than the baseline approach across all three traffic modes, receiving a higher evaluation reward even in the difficult traffic mode. This improvement is attributed to the replacement of most unsafe actions with safe actions by the safety mechanism, particularly during the early exploration phase. This substitution helps avoid early termination and improves learning efficiency. Fig. 9 shows the average vehicle speed during training, which serves as an indicator of traffic throughput. It is clear that the algorithm with the safety mechanism maintains a higher average speed than the baseline approach (i.e., without the safety mechanism). For example, the baseline + safety mechanism (baseline+sm) achieves around 32 m/s in simple traffic mode, while the baseline approach only reaches approximately 26 m/s. Meanwhile, the bi-modal temporal modeling method further enhances training reward and average speed, as shown in Fig. 8. Under moderate and difficult traffic modes, the baseline + safety mechanism + bi-modal temporal modeling (baseline+sm+btm) demonstrates improved learning speed and training reward. It outperforms the baseline + safety mechanism (baseline+sm) due to the time-series module's ability to utilize more historical information to guide the policy network in selecting the best action. Fig. 9 also indicates that the bi-modal temporal modeling method achieves the highest average speed in moderate and difficult traffic modes compared to the method that employs only the safety mechanisms.

After completing the training, we tested three random seeds for each traffic density algorithm over 100 rounds. The resulting collision rate, driving distance, and average speed are shown in Table 4. our method consistently achieved zero collision rates across all traffic scenarios, while PPO without safety reached collision rates of 0.03–0.04 in medium and hard scenarios. This confirms the superiority of our safety mechanism in ensuring collision-free lane changes. The data in Table 4 indicate that the baseline approach without the safety mechanism performs poorly in difficult traffic modes. In contrast, the baseline + safety mechanism (baseline+sm) substantially reduces collisions: it achieves zero collision rate in the Easy and Hard scenarios and reduces the collision rate in the Mid scenario to CR = 0.03. The full system (baseline+sm+btm) further improves safety and attains zero collision rate across all three traffic modes.

Table 4 and the associated training curves explicitly compare (i) the baseline, (ii) baseline + safety mechanism, and (iii) baseline + safety + bi-modal temporal modeling. These comparisons quantify the contributions of the safety module and the temporal module, adding the safety mechanism substantially reduces collision incidence, while incorporating the temporal module further improves training stability and average speed.

In addition to reducing vehicle collision rates, we also evaluated the impact of the Game-based Safety Mechanism on traffic flow. We integrated varying proportions of traditional human-driven vehicles and connected vehicles, establishing three traffic density scenarios corresponding to 0%, 50%, and 100% market penetration rates. A comparative analysis was conducted between the PPO model without the Game-based Safety Mechanism and the PPO model incorporating the mechanism. The experiments were analyzed based on vehicle density and traffic flow, with results demonstrating that the proposed method not only reduces collision rates but also enhances traffic mobility.

Table 4
Comparison of collision rate and average speed.

Models	Easy			Mid			Hard		
	CR	DT (m)	AS (m/s)	CR	DT (m)	AS (m/s)	CR	DT (m)	AS (m/s)
baseline	0	6043.71	25.37	0.03	6010.64	25.23	0.04	5991.52	24.73
baseline+sm	0	6933.01	34.08	0.03	6905.69	28.96	0	6901.28	29.78
baseline+sm+btm	**0**	**6947.25**	**34.15**	**0**	**6943.86**	**31.95**	**0**	**6941.36**	**33.11**

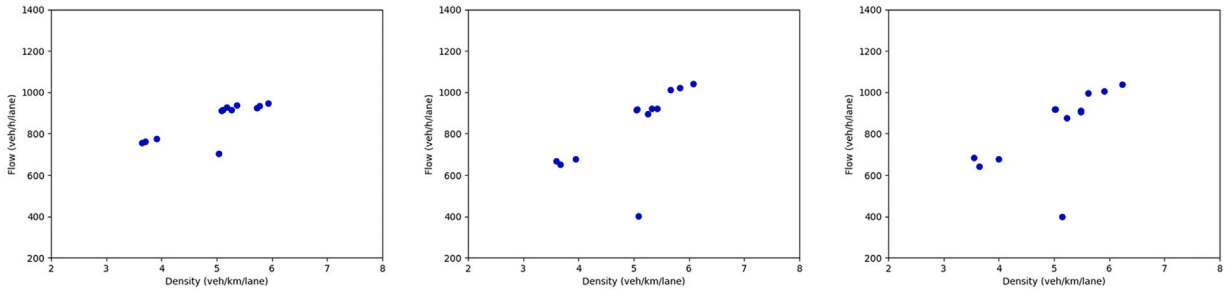

Fig. 10. Lane-changing model without the Game-based Safety Mechanism.

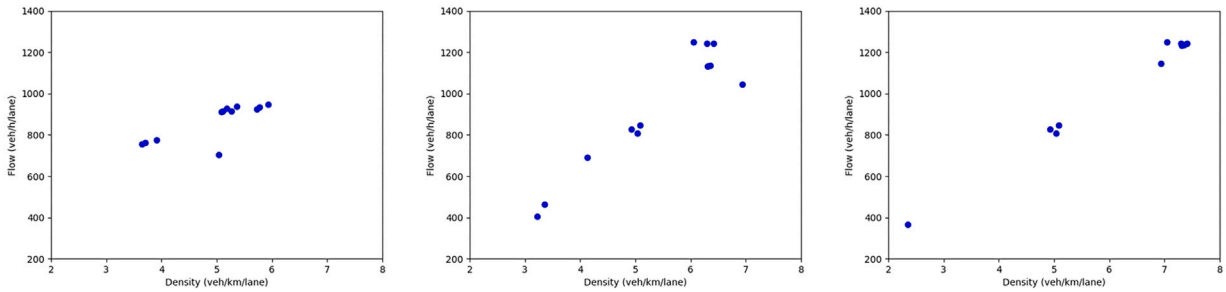

Fig. 11. Lane-changing model with the Game-based Safety Mechanism.

As shown in Figs. 10 and 11, the fundamental diagrams for the models without and with the Game-based Safety Mechanism are displayed under 0%, 50%, and 100% market penetration rates, respectively. The horizontal axis represents the number of vehicles per kilometer per lane, while the vertical axis denotes the number of vehicles passing per hour per lane. When data points cluster in the low-density, low-flow region, it indicates sparse vehicle presence and low traffic flow, typically characteristic of off-peak hours with smooth traffic conditions. However, the flow does not reach its maximum, suggesting under utilization of the road network. When points are distributed in the medium-density, high-flow region, it reflects an optimal traffic state where vehicle volume is moderate, and flow is maximized, indicating efficient traffic system operation with orderly vehicle movement and full utilization of road resources. Conversely, when points concentrate in the high-density, low-flow region, it signifies overcrowding, reduced flow, and the onset of congestion, representing an inefficient state of the traffic system with severe traffic deterioration.

Specifically, as illustrated in Fig. 10, it can be observed that at lower market penetration rates, the model without the Game-based Safety Mechanism exhibits more scattered data points. At a 50% market penetration rate, although the model without the mechanism shows improvements in density and flow in the high-density and high-flow regions, it generates breakdown flow in the medium-density, low-flow region. This phenomenon, where vehicle count increases but flow decreases, leads to a sharp decline in traffic efficiency. Furthermore, at a 100% market penetration rate, the breakdown flow produced by the model without the mechanism shows no significant difference compared to that at the 50% penetration rate.

As demonstrated in Fig. 11, the model with the Game-based Safety Mechanism exhibits a more desirable scatter distribution. For instance, at a 50% market penetration rate, the data points for the model with the mechanism are concentrated in the high-density, high-flow region, indicating optimal traffic system efficiency with maximized vehicle volume and flow, and full utilization of road resources. In contrast to the model without the mechanism, our approach does not produce breakdown flow, which otherwise leads to unrealistic congestion patterns. Moreover, at a 100% market penetration rate, the data points are primarily distributed across low-density low-flow, medium-density medium-flow, and high-density high-flow regions. This distribution signifies good traffic conditions under low vehicle density and flow, moderate traffic volume and optimal efficiency under medium conditions, and maximized road resource utilization under high density and flow. In summary, across all vehicle densities and flow conditions, our method does not generate anomalous breakdown flow, demonstrating that the Game-based Safety Mechanism not only significantly enhances road safety but also achieves coordination among different vehicles without compromising overall performance.

We compared our method to baseline approaches such as PPO, A2C, SAC, and Rainbow DQN. Figs. 12 and 13 show a comparison between the proposed approach and three mainstream benchmarks. As expected, our method demonstrated the best and fastest learning ability, as illustrated in Fig. 12. Under all traffic difficulties, the training curve of our method increased steadily and stabilized in the shortest time. In contrast, Rainbow DQN and SAC maintained better performance only in simple traffic modes; however, in moderate and difficult traffic modes, the complexity of the traffic caused their performance to decline. On the other hand, as shown in Fig. 12, our method achieves the highest average speed compared to other methods in all traffic modes. For example, our method reaches a speed of 32 m/s, compared to 23 m/s for A2C, 26 m/s for SAC, and 28 m/s for Rainbow DQN. This improvement is attributed to the proposed bi-modal temporal modeling

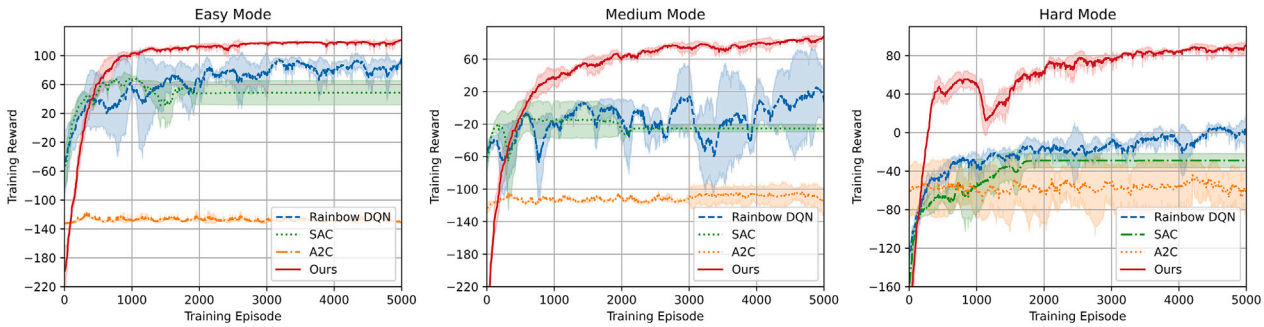

Fig. 12. Training reward comparisons between the three most popular baselines (Rainbow DQN, A2C and SAC) and our method in three traffic modes.

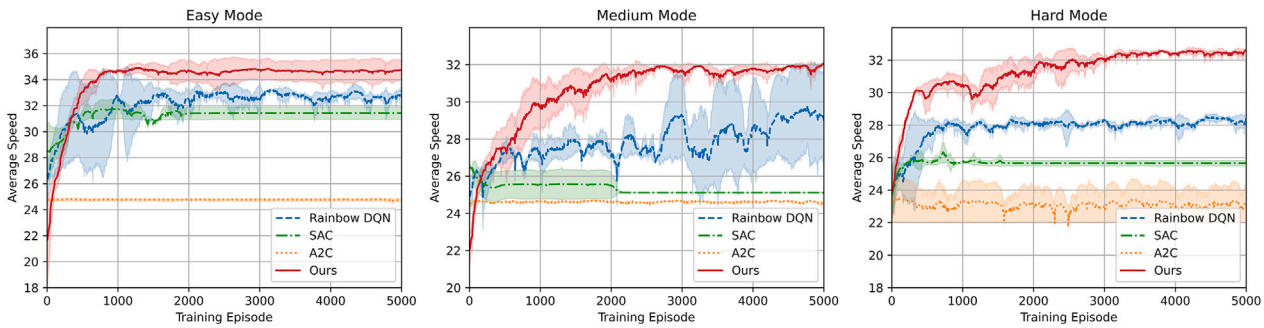

Fig. 13. Average speed comparisons between the three most popular baselines (Rainbow DQN, A2C and SAC) and our method in three traffic modes.

Table 5
Comparison of collision rate and average speed.

Models	Easy			Mid			Hard		
	CR	DT (m)	AS (m/s)	CR	DT (m)	AS (m/s)	CR	DT (m)	AS (m/s)
Rainbow DQN	0	6931.07	33.79	0.04	6344.58	26.62	0	6453.87	27.07
A2C	0.07	5843.89	24.88	0.2	5778.12	24.26	0.14	4752.70	19.99
SAC	0	6654.90	31.69	0.12	5984.69	25.11	0.1	6141.55	25.76
Ours	**0**	**6947.25**	**34.15**	**0**	**6943.86**	**31.95**	**0**	**6941.36**	**33.11**

Table 6
Performance comparison in the ramp merging scenario.

Models	Collision rate	Merge success rate	Avg speed (m/s)
baseline	0.06	0.82	26.4
baseline + sm	0.00	0.90	28.7
baseline + sm + btm	**0.00**	**0.95**	**30.2**

reinforcement learning and the integrated safety mechanism, which enhance sample efficiency and accelerate learning speed.

We tested three random seeds for each algorithm over 100 rounds in each traffic mode. Table 5 presents the test results of the proposed method and the benchmark methods in terms of collision rate, travel distance, and average speed. Clearly, across different traffic modes, our approach consistently achieves the best performance compared to other algorithms. In particular, our method maintains a zero collision rate even in the most challenging situations, specifically in difficult traffic patterns. While Rainbow DQN and SAC without a safety mechanism also demonstrated good performance in simple traffic mode, they struggled in moderate traffic mode, exhibiting high collision rates, and performed poorly in difficult traffic mode. A2C consistently under performed in all scenarios, which aligns with the performance trends shown in the training curves in Figs. 12 and 13.

Table 6 reports the performance of different methods in the ramp merging scenario, evaluated over three random seeds. The results are presented in terms of collision rate, merge success rate, and average speed, which jointly reflect safety and operational efficiency in a realistic highway merging context.

The PPO baseline exhibits a relatively high collision rate and a lower merge success rate, indicating difficulties in handling mandatory merging interactions under priority constraints. This observation is consistent with its performance in dense mainline traffic, where the lack of an explicit safety mechanism leads to unstable decision-making when vehicle interactions intensify. By incorporating the game-based safety mechanism, the baseline+sm approach significantly reduces the collision rate while improving the merge success rate. This suggests that the safety mechanism remains effective in mitigating risky maneuvers even when the driving task shifts from discretionary lane changes to mandatory merging.

The full framework (baseline+sm+btm) achieves the lowest collision rate and the highest merge success rate among all methods. In addition, it maintains a higher average speed in the merging area, indicating that safety improvements are not achieved at the expense of excessive conservatism. These results suggest that the proposed bimodal temporal modeling and safety mechanism do not immediately degrade when transferred to a mandatory merging scenario, indicating a certain degree of robustness beyond the mainline lane-changing setting. It should be emphasized that this ramp merging experiment is intended as a preliminary generalization test rather than a comprehensive evaluation of ramp merging strategies. The scenario is not optimized for ramp-specific behaviors, nor does it account for broader factors such as heterogeneous ramp geometries or perception uncertainty. A detailed investigation of ramp-specific policy optimization and large-scale real-world validation is left for future work. Therefore, the purpose of this experiment is not to claim state-of-the-art performance in ramp merging, but to provide supporting evidence that the learned

policy and safety mechanism are not overfitted to a single mainline lane-changing configuration.

5. Discussion

5.1. Performance interpretation and modeling insights

The research results show that the integration of the proposed bi-modal temporal modeling reinforcement learning algorithm and lane change safety mechanism demonstrates consistent improvements in driving safety and efficiency across a range of highway scenarios.. The task of bi-modal temporal modeling is to model the bi-modal information of the relationship between the environmental state observed by CAV and the vehicle. At the same time, the time-series module is used to capture the time dependence and historical trajectory of traffic interaction so that the agent can better capture potential dangers, so as to help the agent better adapt to complex traffic dynamics, so that the intelligent system can determine a more robust and accurate lane change strategy.

5.2. Limitations and scope of the study

It is worth noting that the time-series module also introduces higher computational costs and delays. Nevertheless, the game-theoretic safety controller adds only a small constant overhead, and the overall inference cycle remains within typical 10–20 Hz control rates in our experiments.

As shown in Figs. 8 and 9, in simple and moderate traffic modes, the training rewards and average driving speeds of the baseline + safety mechanism + bi-modal temporal modeling (baseline+sm+btm) method are lower than those of the baseline + safety mechanism (baseline+sm) method without temporal modeling in the early stages of training. This phenomenon can be attributed to the fact that temporal modeling tends to impose more conservative action selection during early training stages, which may temporarily limit exploratory behavior compared with models without temporal dependencies.

The computational time required for training the agent is acceptable for offline simulations, with training taking approximately 12 h per traffic mode on an NVIDIA RTX 2080Ti. The inference latency increased by approximately 8 ms due to the safety mechanism, but remains within acceptable limits for real-time applications. Future work will explore model pruning and parallelization to further optimize runtime performance for real-time deployment. This reported inference latency includes both the policy network forward pass and the safety mechanism evaluation, ensuring that the overall control loop remains within the stated real-time bounds.

In simple traffic mode, the performance difference between the baseline+sm+btm method and other methods is minimal, as observed in Table 4. Our method also shows limited improvement in driving distance and average speed compared to Rainbow DQN in Table 5. This can be attributed to the relatively straightforward nature of the simple mode, where various algorithms can quickly identify the optimal strategy, leading to no significant performance disparities.

And the game-theoretic Nash equilibrium is computed based on the positions and velocities of the vehicles involved. While the current formulation works well for a small number of vehicles, we plan to scale this mechanism to handle larger numbers of surrounding vehicles. Future work will explore using approximate equilibrium solvers or hierarchical decomposition to maintain computational feasibility in more complex traffic environments. As the difficulty of the traffic mode increases, the performance of other methods declines, particularly in difficult traffic conditions, where our method shows improved robustness to handle more complex scenarios.

5.3. Limitations and practical considerations

While we performed a grid search for reward weights w_c, w_s, w_l and reported sensitivity within $\pm 10\%$, we did not conduct a dedicated sensitivity analysis on the safety-entry threshold X_{safe}, which determines when the game-theoretic controller is activated. We acknowledge that varying X_{safe} and related safety margins is important for evaluating the robustness of both the safety mechanism and policy behavior. Systematic sensitivity sweeps (e.g., $\pm 10\%$ to $\pm 30\%$ around nominal thresholds) will be explored in future extensions of this work.

Likewise, encoder-level ablation studies (e.g., bi-modal versus ego-only or relational-only representations) and additional robustness evaluations, including cross-map generalization, sensor noise and occlusion, delayed actions, and diverse HDV behavior mixes, are not included in the present manuscript. This is because the current study focuses on a single-agent PPO-based decision-making framework within controlled SUMO highway scenarios. These limitations define the current scope of the study and point to important directions for future research.

5.4. Practical implications

The findings from our study on the bi-modal temporal modeling reinforcement learning algorithm for CAV lane-changing behavior have significant practical implications across various domains, including traffic management, vehicle design, and policy-making.

The improved lane-changing efficiency and reduced collision rates exhibited by our model may offer insights for the development of future traffic management systems under controlled deployment conditions. By integrating our algorithm into real-time traffic control systems, cities can optimize traffic flow, minimize congestion, and enhance overall road safety. Such systems could utilize data from CAVs to dynamically adjust traffic signals, provide real-time lane change guidance, and manage traffic patterns more effectively.

These findings underscore the importance of incorporating advanced decision-making algorithms into vehicle design, particularly for CAVs. Manufacturers can leverage the proposed framework to enhance onboard decision-making systems, enabling vehicles to adapt more effectively to complex traffic conditions. In parallel, policy initiatives that promote smart infrastructure and CAV-supportive environments may further facilitate the safe deployment of automated driving technologies.

Overall, the practical implications of this work highlight the potential of reinforcement learning-based lane-changing strategies to contribute to safer and more efficient transportation systems, while also emphasizing the importance of addressing real-world constraints in future research.

6. Conclusion

In this paper, we proposed a bi-modal temporal modeling reinforcement learning framework with an integrated game-theoretic safety mechanism for Connected and Autonomous Vehicles (CAVs) in mixed highway traffic. By jointly modeling ego-state information and inter-vehicle relationships, incorporating temporal dependencies of historical trajectories, and dynamically adapting to surrounding vehicles through Nash equilibrium computations, the proposed method enables safer and more efficient lane-changing decisions.

The proposed approach was evaluated using the SUMO microscopic traffic simulator under diverse traffic densities and driving styles. The experimental results indicate that the framework consistently reduces collision rates, improves average vehicle speed, and enhances overall traffic throughput when compared with state-of-the-art reinforcement learning baselines (e.g., Rainbow DQN, A2C, and SAC), as well as PPO variants without the safety or temporal modeling components. These results support the effectiveness of the three main contributions summarized in Section 1: (1) bi-modal state modeling of CAVs and

surrounding vehicle interactions, (2) temporal integration of historical trajectory information, and (3) the incorporation of a game-theoretic safety mechanism for collision avoidance.

Despite these promising results, several limitations of the present study should be acknowledged. The current experimental evaluation is restricted to highway lane-changing scenarios and a preliminary on-ramp merging setting which was introduced to assess generalization rather than to optimize ramp-specific behaviors and does not yet encompass more complex urban driving environments. In addition, the robustness of the proposed framework under practical challenges such as sensor noise, occlusion, and delayed actions has not been systematically assessed. Furthermore, comfort- and fairness-related metrics, including lateral and longitudinal jerk, lane-change duration, and the forced-braking rate of surrounding vehicles, are not explicitly analyzed in this work.

These limitations define the current scope of the study and motivate several directions for future research. Future work will extend the proposed framework to more complex urban driving scenarios, such as intersections, roundabouts, and merging areas, as well as high-density mixed traffic conditions. In addition, robustness and comfort-related metrics will be incorporated to provide a more comprehensive evaluation. We also plan to investigate multi-agent and transformer-based reinforcement learning extensions to further enhance adaptability and scalability in complex traffic environments.

CRediT authorship contribution statement

Xing Xu: Writing – original draft, Visualization, Methodology, Investigation. **Tingpeng Shi:** Writing – original draft, Methodology, Investigation. **Zhang Zhang:** Writing – review & editing, Methodology, Conceptualization. **Yun Zhao:** Supervision, Methodology, Conceptualization. **Yali Wang:** Writing – review & editing. **Yanan Mi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Natural Science Foundation of China (62476251); National Key Research and Development Program of China (2019YFE0126100).

Data availability

Data will be made available on request.

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